Compiled Solutions – Intermediate Division

1. **00-01 C3 Digital Electronics**

The circuit translates as follows: (1,1,1), (0,1,1)
$\overline{A+B}A+\overline{BC}=(\overline{A}*\overline{B})*A+\overline{BC}=0+\overline{BC}=\overline{BC}$
This is false if BC is true. Therefore, the answer is (*,1,1).

2. **00-01 C3 Digital Electronics**

The circuit translates as follows: As shown
$\overline{\overline{AB}\oplus\overline{A}+\overline{B\oplus C}}$

3. **00-01 C4 Digital Electronics**

The truth table forms as follows: 4

A	B	C	$\overline{A+B}$	$\overline{BC}$	AND
0	0	0	1	1	1
0	0	1	1	1	1
0	1	0	1	1	1
0	1	1	1	0	0
1	0	0	0	1	0
1	0	1	0	1	0
1	1	0	1	1	1
1	1	1	1	0	0

4. **01-02 C3 Digital Electronics**

The circuit translates as follows: $(A+B)\overline{B}$.
To be TRUE both factors must be TRUE. B must be equal to 0. (1,0)
Therefore, A must be equal to 1.

5. **01-02 C3 Digital Electronics**

The circuit translates as follows: $(\overline{XY}\oplus\overline{Z}$. It simplifies to:
$(X+\overline{Y})(YZ+\overline{Y}\ \overline{Z})=XYZ+X\ \overline{Y}\ \overline{Z}+\overline{Y}\ Y\ Z+\overline{Y}\ \overline{Y}\ \overline{Z}=$ $XYZ+\overline{Y}\ \overline{Z}$
$XYZ+X\ \overline{Y}\ \overline{Z}+\overline{Y}\ \overline{Z}=XYZ+\overline{Y}\ \overline{Z}(X+1)=XYZ+\overline{Y}\ \overline{Z}$

6. **01-02 C4 Digital Electronics**

The circuit translates to A(B + B). This simplifies to A. D
The circuit is TRUE for (1, 0) and (1, 1) and FALSE for (0, 1) and (0, 0).

7. **02-03 C3 Digital Electronics**

The circuit translates as follows: (1,0,1)

$A(A\overline{B} \oplus \overline{B+C}) = A((A\overline{B})(\overline{\overline{B+C}}) + (\overline{A\overline{B}})(\overline{B+C})) =$

$A(A\overline{B})(B+C) + A(A\overline{B})(B+C) + A(\overline{A}+B)(\overline{B}\,\overline{C}) =$

$A\overline{B}B + A\overline{B}C + A(\overline{A}\,\overline{B}\,\overline{C}) + A(B\overline{B}\,\overline{C}) = 0 + A\overline{B}C + 0 + 0 =$

$A\overline{B}C$ so A=1, B=0, and C=1.

8. **02-03 C3 Digital Electronics**

Using the fewest sets of parentheses, the circuit translates to: $\overline{\overline{A\overline{B}}(B \oplus C)(C+D)D}$

$\overline{\overline{A\overline{B}}(B \oplus C)(C+D)D}$

The operands may be commuted.

9. **02-03 C4 Digital Electronics**

$\overline{\overline{A+\overline{B}}(BC)} = 0$ implies $\overline{A+\overline{B}}(BC) = 1$ 0

If BC = 1, then B = 1 and C = 1, if A+B = 1, then A = 0 and B=0.
This is a contradiction since B must be 0 in one case and 1 in the other case.
Our implied case is always FALSE so the original statement is always TRUE.

10. 03-04 C3 Digital Electronics

The circuit translates to: $\overline{\overline{A}(B+C)+C}$

A	B	C	$B+C$	$\overline{A}$	$\overline{A}(B+C)$	$\overline{A}(B+C)+C$	NOT
0	0	0	0	1	0	0	1
0	0	1	1	1	1	1	0
0	1	0	1	1	1	1	0
0	1	1	1	1	1	1	0
1	0	0	0	0	0	0	1
1	0	1	1	0	0	1	0
1	1	0	1	0	0	0	1
1	1	1	1	0	0	1	0

11. 03-04 C3 Digital Electronics

The circuit translates to: $(\overline{A}+AB)(B+C)\overline{C}$ $B\overline{C}$

$= (\overline{A}+AB)(B\overline{C}+0) = (\overline{A}+AB)B\overline{C} = \overline{A}B\overline{C} + AB\overline{C}$

$= B\overline{C}(\overline{A}+A) = B\overline{C}(1) = B\overline{C}$

12. 03-04 C4 Digital Electronics

The circuit converts to Boolean form and simplifies as follows: (*,0,1), (0,1,*)

$(\overline{A}+\overline{AB})\oplus\overline{C+B\overline{C}}=(\overline{A}+\overline{A}+\overline{B})\oplus(\overline{C}*\overline{B\overline{C}})=$ OR

$(\overline{A}+\overline{B})\oplus(\overline{C}(\overline{B}+\overline{\overline{C}}))=\overline{AB}\oplus(\overline{B}\overline{C}+C\overline{C})$ (1,0,1), (0,0,1),

$\overline{AB}\oplus\overline{B}\overline{C}$ must be TRUE (0,1,1), (0,1,0)

Suppose $\overline{AB}=0$ and $\overline{B}\overline{C}=1$. This leads to an impossible situation.

Now suppose $\overline{AB}=1$ and $\overline{B}\overline{C}=0$

If B=0 then A=* and C=1

If B=1 then A=0 and C=*

13. 04-05 C3 Digital Electronics

The circuit translates to: $\overline{AB+(B\oplus C)}=1$. (1,0,0), (0,0,0), (0,1,1)

$\Rightarrow AB+(B\oplus C)=0.$

$\Rightarrow AB=0$ AND $B\oplus C=0$

If $B=0\Rightarrow A=*, C=0$

If $B=1\Rightarrow A=0, C=1$

14. 04-05 C3 Digital Electronics

With the minimum number of parentheses required the circuit translates as follows: $\overline{\overline{A}B(\overline{B+C})\oplus C}$. $\overline{\overline{A}B(\overline{B+C})\oplus C}$

15. 04-05 C4 Digital Electronics

The circuit translates to: $(AB\oplus\overline{B+C})\overline{C}$. (0,0,0), (1,0,0), (1,1,0)

A	B	C	AB	$\overline{B+C}$	$\oplus$	$\overline{C}$	*
0	0	0	0	1	1	1	1
0	0	1	0	0	0	0	0
0	1	0	0	0	0	1	0
0	1	1	0	0	0	0	0
1	0	0	0	1	1	1	1
1	0	1	0	0	0	0	0
1	1	0	1	0	1	1	1
1	1	1	1	0	1	0	0

16. 05-06 C3 Digital Electronics

The truth table ;looks like: 4

A	B	C	AB	$\overline{B+C}$	$\oplus$
0	0	0	0	1	1
0	0	1	0	0	0
0	1	0	0	0	0
0	1	1	0	0	0
1	0	0	0	1	1
1	0	1	0	0	0
1	1	0	1	0	1
1	1	1	1	0	1

17. 05-06 C3 Digital Electronics

$\overline{\overline{A}\overline{B}+(B+C)} = \overline{\overline{A}\overline{B}}(\overline{B+C}) = (A+B)(\overline{B}\overline{C}) =$ $A\overline{B}\overline{C}$

$A\overline{B}\overline{C} + B\overline{B}\overline{C} = A\overline{B}\overline{C}$

18. 05-06 C4 Digital Electronics

The circuit translates to $(\overline{A} \oplus AB)((B+C)C) = 1$. C must = 1. (1,1,1), (0,1,1),

If $B = 1$ then the expression simplifies to $\overline{A} \oplus A = 1$. This means that A = *. (0,0,1)

If $B = 0$ then the expression simplifies to $\overline{A} \oplus 0 = 1$. This means that $A = 0$. The ordered triplets are (*,1,1) and (0,0,1).

19. 06-07 C3 Digital Electronics

The circuit simplifies to $\overline{\overline{A}B + \overline{BC}} = \overline{\overline{A}B} * \overline{\overline{BC}} = (A+\overline{B})(BC) =$ ABC

$ABC + \overline{B}BC = ABC + 0 = ABC$

20. 06-07 C3 Digital Electronics

The circuit translates to $\overline{\overline{A}(A+B)} + \overline{B\overline{C}+D} = 0$ 3

$\overline{\overline{A}A + \overline{A}B} + \overline{B\overline{C}}\,\overline{D} = 0$ $\qquad (\overline{\overline{A}A})(\overline{\overline{A}B}) + \overline{B\overline{C}}\,\overline{D} = 0$

$\overline{\overline{A}B} + (\overline{B}+C)\overline{D} = 0$ $\qquad A + \overline{B} + \overline{B}\overline{D} + C\overline{D} = 0$

$\Rightarrow$ all 4 terms must be 0.

If D = 0 $\Rightarrow$ C = 0 $\Rightarrow$ (0,1,0,0)

If D = 1 $\Rightarrow$ C = * $\Rightarrow$ (0,1,*,1)

21. 06-07 C4 Digital Electronics

The digital diagram translates as follows: $\overline{A}B\overline{C}$

$\overline{A}(AB + (B + C)\overline{C}) =$

$\overline{A}(AB + B\overline{C} + C\overline{C}) = \overline{A}AB + \overline{A}B\overline{C} + \overline{A}C\overline{C} = \overline{A}B\overline{C}$

22. 07-08 C3 Digital Electronics

The circuit translates to $(\overline{A}((\overline{A + B})\overline{BC})) \oplus \overline{BC}$ (0,0,0), (0,0,1)

$(\overline{A}((\overline{A + B})\overline{BC})) \oplus \overline{BC} = (\overline{A}(\overline{A}\,\overline{B}\,\overline{BC})) \oplus \overline{BC} = \overline{A}\overline{B} \oplus \overline{BC}$ (1,1,1), (0,1,1)

To be FALSE, both sides of $\oplus$ must the same

If both T, then (0,0,*). If both F, then (*,1,1)

23. 07-08 C3 Digital Electronics

This circuit translates to $(A(\overline{A + B}))(B\overline{C} + C)$ 0

$= (A\overline{A}\overline{B})(B\overline{C} + C) = (0)(B\overline{C} + C) = 0$

24. 07-08 C4 Digital Electronics

The digital diagram translates as follows: (1,1,0)

$A(\overline{\overline{AB} + \overline{(B + C)\overline{C}}}) = A(\overline{\overline{AB}}\,(\overline{\overline{(B + C)\overline{C}}}))$

$= A((AB(B\overline{C} + C\overline{C}))$

$= AB\overline{C} + ABC\overline{C} = AB\overline{C} + 0 = AB\overline{C} = 1$

$AB\overline{C} = 1 \Rightarrow A = 1,\ B = 1,$ and $C = 0$ $\therefore$ (1, 1, 0)

25. 08-09 C3 Digital Electronics

The circuit translates to $\overline{AB \oplus \overline{BC}}$. 2

A	B	C	AB	$\overline{BC}$	$\overline{AB \oplus \overline{BC}}$
0	0	0	0	1	0
0	0	1	0	1	0
0	1	0	0	1	0
0	1	1	0	0	1
1	0	0	0	1	0
1	0	1	0	1	0
1	1	0	1	1	1
1	1	1	1	0	0

26. 08-09 C3 Digital Electronics

This circuit translates to: $((\overline{A}(\overline{B+C}))(C+\overline{D})$ $\overline{A}\overline{B}\overline{C}\overline{D}$

$$= \overline{A}\,\overline{B}\,\overline{C}(C+\overline{D}) = \overline{A}\,\overline{B}\,\overline{C}\,C + \overline{A}\,\overline{B}\,\overline{C}\,\overline{D}$$
$$= 0 + \overline{A}\,\overline{B}\,\overline{C}\,\overline{D} = \overline{A}\,\overline{B}\,\overline{C}\,\overline{D}$$

27. 08-09 C4 Digital Electronics

The digital diagram translates as follows: (1,1,0), (0,1,0)

$\overline{A\overline{B} + (B \oplus \overline{C})} = 1 \Rightarrow A\overline{B} + (B \oplus \overline{C}) = 0 \Rightarrow A\overline{B} = 0 \wedge (B \oplus \overline{C}) = 0$ (0,0,1)

If $B = 1$, then $A = * \wedge C = 0$

If $B = 0$, then $A = 0 \wedge C = 1$

$\therefore$ (1,1,0) (0,1,0) (0,0,1)

28. 09-10 C3 Digital Electronics

The digital diagram translates as follows: (0,0,0)

$$(\overline{\overline{A+B} \oplus \overline{BC}})\overline{C} = ((A+B)(BC) + (\overline{A+B})(\overline{BC}))\overline{C}$$
$$= (ABC + BC + \overline{A}\overline{B}(\overline{B}+\overline{C}))\overline{C}$$
$$= ABC\overline{C} + BC\overline{C} + \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}\overline{C} = 0 + 0 + \overline{A}\overline{B}\overline{C}$$

29. 09-10 C3 Digital Electronics

The circuit translates as follows: $(\overline{A} \oplus AB) \square (\overline{B+C}) \square \overline{C}$ (0,1,1), (1,0,1), (1,1,1)

A	B	C	$\overline{A}$	AB	$\overline{A} \oplus AB$	$B+C$	$\overline{B+C}$	$\overline{C}$	$\square$
0	0	0	1	0	1	0	1	1	0
0	0	1	1	0	1	1	1	0	0
0	1	0	1	0	1	1	1	1	0
0	1	1	1	0	1	1	0	0	1
1	0	0	0	0	0	0	1	1	0
1	0	1	0	0	0	1	1	0	1
1	1	0	0	1	1	1	1	1	0
1	1	1	0	1	1	1	0	0	1

30. 09-10 C4 Digital Electronics

The digital diagram translates as follows:

$(\overline{AB} \oplus (B+C))C = 1 \Rightarrow (\overline{AB} \oplus (B+C)) = 1 \wedge C = 1$ (1,1,1)

$$\Rightarrow (\overline{AB} \oplus (B+1))1 = 1$$
$$\Rightarrow \overline{AB} = 0 \Rightarrow AB = 1 \Rightarrow A = 1 \wedge B = 1 = 1$$

$\therefore$ (1, 1, 1)

31. 10-11 C3 Digital Electronics

The circuit translates to $\overline{A}(\overline{\overline{A}+B})(\overline{BC})$ 2

$\overline{A}(\overline{\overline{A}+B})(\overline{BC}) = (\overline{A}(\overline{A}\overline{B}))(\overline{B}+\overline{C}) = (\overline{A}\overline{B}))(\overline{B}+\overline{C})$

$= \overline{A}\overline{B} + \overline{A}\overline{B}\overline{C} = \overline{A}\overline{B}(1+\overline{C}) = \overline{A}\overline{B}$

If $\overline{A}\overline{B}=1$, then $\overline{A}=1 \wedge \overline{B}=1$. So $A=0 \wedge B=0 \wedge C=*$.

Therefore there are 2 ordered pairs that make the circuit TRUE.

32. 10-11 C3 Digital Electronics

This circuit translates to $(\overline{AB})(\overline{(B+C)+\overline{CD}})+D$ D

$(\overline{AB})(\overline{(B+C)+\overline{CD}})+D = (\overline{A}+\overline{B})(\overline{B}\overline{C}CD)+D = (\overline{A}+\overline{B})(0)+D = D$

33. 10-11 C4 Digital Electronics

The digital diagram translates as follows: 2

$(\overline{\overline{A}+AB})(\overline{\overline{B+C}}\overline{C}) = (\overline{\overline{A}}\overline{AB})(\overline{\overline{B+C}+\overline{C}}) = (A(\overline{A}+\overline{B}))((B+C)+\overline{C})$

$= (A\overline{A}+A\overline{B})(B+1) = (0+A\overline{B})(1) = A\overline{B}.$

$\therefore$ The circuit will be TRUE when the solution is combined with C and NOT C.

34. 11-12 C4 Digital Electronics

$(A+\overline{AB})((B+C)C) = (A+\overline{A}+\overline{B})(BC+C) = (1+\overline{B})(C(B+1)) = C$ C

35. 11-12 C4 Digital Electronics

The circuit translates to: (0,0,0), (0,1,0), (1,1,0)

$\overline{A(\overline{B+C})+C} = 1.$

Constructing a truth table will show that the circuit is true for
(0, 0, 0), (0, 1, 0) and (1,1, 0).

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36. 12-13 C4 Digital Electronics

The circuit translates to: $\overline{\overline{AB}+(B+C)}$ NONE

$\overline{\overline{AB}+(B+C)} = \overline{\overline{AB}} * \overline{B+C} = AB\overline{B}\overline{C} = 0$

Therefore no triples make the circuit true.

37. 12-13 C4 Digital Electronics

The circuit translates to: $\overline{((A+\overline{A}B)+\overline{B\overline{C}})\overline{C}}$ 0

$\overline{((A+\overline{A}B)+\overline{B\overline{C}})\overline{C}}=1\Rightarrow \overline{\mathrm{C}}=1\Rightarrow C=0$

$\overline{((A+\overline{A}B)+\overline{B\overline{C}})}=1\wedge \mathrm{C}=0\Rightarrow \mathrm{A}+\overline{\mathrm{A}}\mathrm{B}+\overline{\mathrm{B}0}=1\Rightarrow A+\overline{A}B+1=0$

which is impossible. Therefore no ordered triples make it true.

38. 13-14 C4 Digital Electronics

The circuit translates to: $\overline{(\overline{\overline{A}+BC})\overline{C}}$ (1,1,0), (1,0,0)

$\overline{(\overline{\overline{A}+BC})\overline{C}}=\overline{(\overline{\overline{A}+BC})}+\overline{\overline{C}}=(\overline{A}+BC)+C=\overline{A}+C$

$\overline{A}+C=0\rightarrow \overline{A}=0\wedge C=0$ Therefore (1, *, 0) makes it FALSE.

39. 13-14 C4 Digital Electronics

The circuit translates to: $A[\overline{(\overline{A\overline{B}}(B+C))+\overline{C}}]$ $A\overline{B}C$

$A[\overline{(\overline{A\overline{B}}(B+C))+\overline{C}}]=A[\overline{(\overline{A\overline{B}}(B+C))}\overline{\overline{C}}]=A[(\overline{\overline{A\overline{B}}}+\overline{(B+C)})C]$

$AC(A\overline{B}+\overline{B}\overline{C})=ACA\overline{B}+AC\overline{B}\overline{C}=A\overline{B}C$

40. 14-15 C4 Digital Electronics

The circuit translates as follows: $\overline{A}\overline{B}C$

$(\overline{A\,B})(\overline{A+B})(C)=(\overline{A}+\overline{B})(\overline{A}\,\overline{B})(C)$

$=(\overline{A}\,\overline{B}+\overline{A}\,\overline{B})(C)=\overline{A}\,\overline{B}C$

41. 14-15 C4 Digital Electronics

The circuit translates as follows: (1,1,1)

$\overline{(\overline{A}+\overline{A\,B})+(B\,\overline{C})}=\overline{(\overline{A}+\overline{A\,B})}\,\overline{(B\,\overline{C})}=$

$A\,(A\,B)(\overline{B}+C)=A\,B\,\overline{B}+A\,B\,C=A\,B\,C$

42. 15-16 C4 Digital Electronics

The circuit translates to: $\overline{A\,\overline{AB}+(B+C)\,C}$ 8

$\overline{A\,\overline{AB}+(B+C)\,C} = \overline{(A\overline{AB})}\,\overline{(B+C)}C = (\overline{A}+\overline{\overline{AB}})(\overline{B}\,\overline{C})C$

$= (\overline{A}+AB)(\overline{B}\,\overline{C}C) = (\overline{A}+AB)0 = 0$

All the ordered triples make the circuit FALSE.

43. 15-16 C4 Digital Electronics

The circuit translates to: $(\overline{\overline{A}(A+B)})(\overline{\overline{BC}+\overline{C}})$ ABC

$=(\overline{\overline{A}}+(\overline{A+B}))\overline{\overline{BC}}\,\overline{\overline{C}} = (A+\overline{A}\,\overline{B})BCC$

$ABCC+\overline{A}\,\overline{B}BCC = ABC+0 = ABC$